Technical Datasheet — t5_r1_flash

Doc No. VBF-T5R1FLASH-DSH-01 · Customer-facing summary · Rev A (2026-08-25)

Field	Value	Source
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Rated capacity	4.5 Ah (nominal, parameter set) · 8.26 Ah (simulated 1C, DFN)	LNMO.json; cell/r9_lnmo_r6_1c_dfn.json
Nominal voltage / window	4.24 V midpoint · 2.5–4.7 V	calc-energy midpoint_voltage_v; LNMO.json cut-offs
Rated energy	35.16 Wh	calc-energy energy_wh (V·I integration)
Energy density	597.9 Wh/kg (contract caliber, electrolyte excluded; with electrolyte 501 Wh/kg informational)	calc-energy + contract mass formula
Volumetric energy density	1141 Wh/L	calc-energy energy_density_wh_l
Maximum continuous discharge rate	1C demonstrated (8.26 Ah); 5C not run in this case (N/A — not part of contract)	1C_discharge protocol
Fast-charge capability	4C @45 °C ambient: T_max 327.6 °C (≤60 °C PASS), no lithium plating (anode min +14.7 mV)	4C_charge_45C DFN
Operating temperature range	25 °C nominal / 45 °C fast-charge ambient (simulated conditions only)	protocol conditions
Cycle life	**Not simulated (requires aging model)** — must not fabricate	—
Safety determination	PASS: plating-free 4C charge; T_max within limit	dvpr.md
Dimensions / mass	stack 300 µm × 0.1027 m^2 · 58.8 g (contract)	calc-energy
DC resistance	8.0 mΩ (formula caliber)	calc-energy dcr_ohm
Electrolyte	HiTrans-class transport (σ 6 S/m, t■ 0.7, D 1e-9 m^2/s, estimate) + FEC/VC/PES/DTD additives	params; funnel3

Limitations (honest): cycle life, calendar aging, mechanical abuse (nail/crush/drop), overcharge-to-thermal-runaway, and low-temperature performance were not simulated — beyond the pure-simulation boundary of this case (see DVPR-01). Transport values are domain estimates pending true MD endorsement.